

Theoretical Astrophysics I: Stellar Structure and Evolution

Lecture 7: Simple models of stellar structure

Petri Käpylä, Ivan Milić
pkapyla,milic@leibniz-kis.de

Institut für Sonnenphysik - KIS, Freiburg

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Recap

- ▶ Now, we understand the main processes that govern stellar structure (*not yet evolution!*).
- ▶ To recap our main assumptions:
- ▶ Stars are in hydrostatic equilibrium.
- ▶ Energy is transported either by radiation or convection
- ▶ Hopefully we understand equation of state (connection between T, ρ, P)
- ▶ Hopefully we can calculate the opacity from there
- ▶ And hopefully we can calculate the rate of energy production.

Equations of stellar structure

Here they are, written in terms of r as independent variable:

$$\frac{dP}{dr} = -\rho \frac{Gm}{r^2} \quad (1)$$

$$\frac{dm}{dr} = 4\pi r^2 \rho \quad (2)$$

$$\frac{dT}{dr} = -\frac{3}{4ac} \frac{\kappa \rho}{T^3} \frac{F}{4\pi r^2} \quad (3)$$

$$\frac{dF}{dr} = 4\pi r^2 \rho q \quad (4)$$

$$P = \frac{\mathcal{R}}{\mu_I} \rho T + P_e + \frac{1}{3} a T^4 \quad (5)$$

$$\kappa = \kappa_0 \rho^a T^b \quad (6)$$

$$q = q_0 \rho^m T^n \quad (7)$$

Equations of stellar structure

Count the unknowns, count the equations, is the system, in principle, solvable?

$$\frac{dP}{dr} = -\rho \frac{Gm}{r^2}$$

$$\frac{dm}{dr} = 4\pi r^2 \rho$$

$$\frac{dT}{dr} = -\frac{3}{4ac} \frac{\kappa \rho}{T^3} \frac{F}{4\pi r^2}$$

$$\frac{dF}{dr} = 4\pi r^2 \rho q$$

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Equations of stellar structure

Four of these equations are differential equations. Three are algebraic. Differential equations are of the first order, so each one needs one boundary condition. These are, generally speaking.

- ▶ At $r = 0$ (center), $m = 0$, $F = 0$.
- ▶ At $m = M$, $P = 0$ and $F = L$.

It turns out that two physical parameters that govern the structure a so-called ZAMS (zero age main sequence star) star are the **total mass** and **chemical composition**. More about that in the next class.

Can someone tell me why we could not say at $r = R$, $m = M$ and $P = 0$ and be done with it?

Equations of stellar structure

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Can someone tell me why we could not say at $r = R$, $m = M$ and $P = 0$ and be done with it? Because we do not know what R is. When solving the equations of stellar structure, we don't know in advance how big the star will be!

General assumptions

To ease our job, some simple things can, generally be assumed

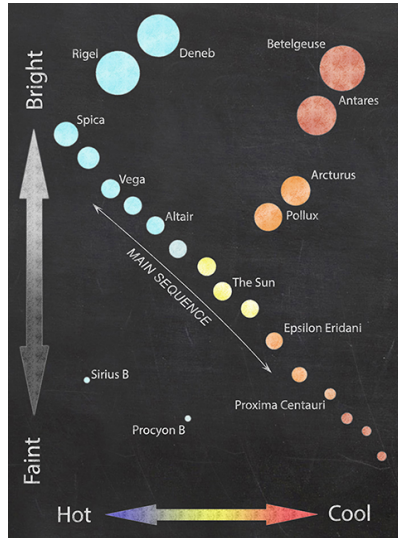
- ▶ Constant μ , either for ZAMS or completely evolved stars.
- ▶ When $q =$, F is constant.
- ▶ P and ρ and T can all be safely assumed to monotonously decrease with r (m).
- ▶ It is super helpful if we can find combinations of quantities that stay approximately constant over the stellar radius.

Now - be mindful that the situation can be even more dire - as the opacity and nuclear rate production can be very complicated and no analytical solution is possible.

In modeling, in general, we solve everything by numerics.

Our end goal

- ▶ Ideally, if we manage to calculate the stellar model, and repeat the calculation for the models of different mass and luminosity, we should get something like HR diagram.
- ▶ It should also give us some very basic insights into the stellar evolution, even if we do not numerically evolve the star.
- ▶ Remember once again, that equations of stellar structure, in a way, tell us what kinds of stars are permitted to exist by physical laws!



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Further simplifications

These equations were written more than 100 years ago. Back then we could not calculate anything numerically (well, almost anything). Now, let's look at the differential equations again:

$$\frac{dP}{dr} = -\rho \frac{Gm}{r^2}$$

$$\frac{dm}{dr} = 4\pi r^2 \rho$$

$$\frac{dT}{dr} = -\frac{3}{4ac} \frac{\kappa \rho}{T^3} \frac{F}{4\pi r^2}$$

$$\frac{dF}{dr} = 4\pi r^2 \rho q$$

(8)

First two equations only involve m, P, ρ . What would I need to close the system?

Further simplifications

What would I need to close the system?

Equation of state! But such an equation of state, where T does not appear, that is where P depends only on ρ (μ can also appear as it is a given constant, but discuss this for a sec).

$$\begin{aligned}\frac{dP}{dr} &= -\rho \frac{Gm}{r^2} \\ \frac{dm}{dr} &= 4\pi r^2 \rho \\ P &= K\rho^\gamma\end{aligned}\tag{9}$$

Where we typically define n as $\gamma = 1 + 1/n$.

So, now we have closed the system of equations, and we can go to town! :-)

The ensuing models are called **polytropes**, and can be solved analytically only in a handful of cases (for specific n 's) - as you have seen in your homework.

Sketching the solution

What would I need to close the system?

$$\begin{aligned}\frac{dP}{dr} &= -\rho \frac{Gm}{r^2} \\ \frac{dm}{dr} &= 4\pi r^2 \rho\end{aligned}\tag{10}$$

Multiply first equation by $\frac{r^2}{\rho}$, differentiate with respect to dr and substitute into the equation two, you will get:

$$\frac{1}{r^2} \frac{d}{dr} \left(\frac{r^2}{\rho} \frac{dP}{dr} \right) = -4\pi G \rho\tag{11}$$

And good old: $P = K\rho^\gamma$